Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

SEMESTER 1 EXAMINATIONS 2025/2026

MODULE: CSC1042 - Computer Networks 2

PROGRAMME(S):

CASE	BSc Computer Apps (Sft.Eng.)
COMSCI	BSc Computer Science
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Program EC

STAGE: 1,3

EXAMINER(S):

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TIME ALLOWED: 2 Hours

INSTRUCTIONS: **Answer all questions.**

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

QUESTION 1 ***[TOTAL MARKS: 28]***

Q 1(a) **[20 Marks]**

Given the IP address 201.50.1.10, you are asked to subnet the network into 8 subnets. Answer the following questions:

1. What is the network IP address to which the given IP belongs?
2. What is the new subnet mask after subnetting?
3. How many bits need to be borrowed to create 8 subnets?
4. What are the network addresses of the resulting subnets?
5. What is the broadcast address of each subnet?
6. What are the first and last usable host IP addresses in each subnet?

Show all your work clearly and use full 4 bytes decimal-dot-notation throughout.

Q 1(b) **[8 Marks]**

Given the network address 172.70.0.0/14, answer the following:

1. How many networks are included in this supernet?
2. What are their IP addresses?
3. What is the broadcast address of this supernet?
4. What is the first and last valid host IP addresses in this supernet?

[End of Question 1]

QUESTION 2 ***[TOTAL MARKS: 29]***

Q 2(a) **[10 Marks]**

Explain how a router creates its routing table using a link state routing approach.

Q 2(b) **[5 Marks]**

Abbreviate the following IPv6 address:

00fc:2900:ea31:0000:cbaf:0000:0000:005d

Q 2(c) **[8 Marks]**

Write the lines to be added to the SAO record to create the following:

1. Type A record that maps the domain name “modules.net” to the IP address 128.30.0.1
2. Type CNAME record for “projects” that points to “modules.net”
3. Type NS record for “ns”, the nameserver responsible for resolving DNS queries of “modules.com”
4. Type PTR record to enable reverse DNS lookup for “books.net”

Q 2(d) **[6 Marks]**

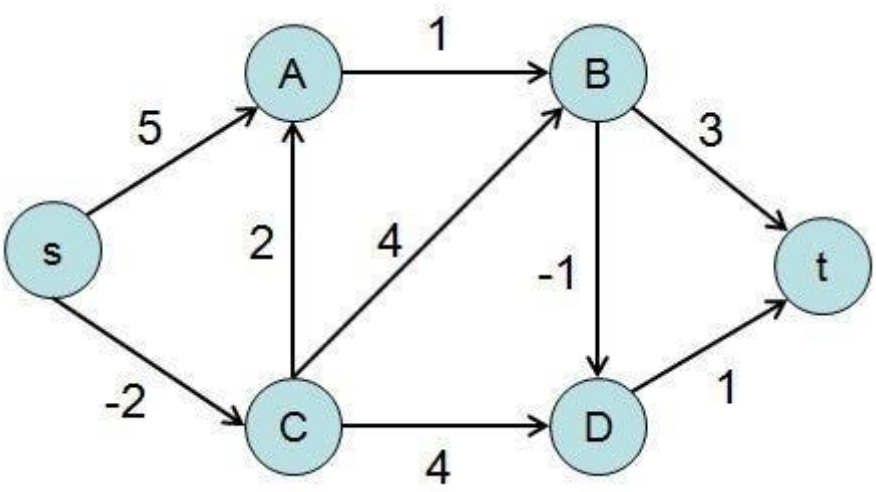
Given a private network with hosts 172.18.3.1 and 172.18.3.2 both communicating with the public IP 25.8.3.2, draw and explain the **NAT translation table** entries that would be created.

[End of Question 2]

QUESTION 3 ***[TOTAL MARKS: 25]***

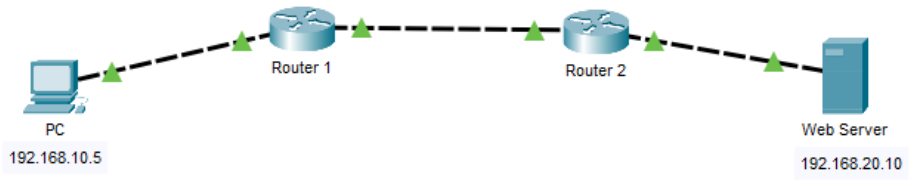
Q 3(a) **[8 Marks]**

Using the graph below, apply the Bellman-Ford algorithm to determine the shortest paths from source node s to all other nodes in the graph. Show all iterations of the algorithm



Q 3(b) **[7 Marks]**

Given the network depicted in the figure below, answer the following:



1. How many networks are present in the topology?
2. Assume the PC wants to send an HTTP request to the Web Server to access a web page. Explain in detail how the Address Resolution Protocol (ARP) process is carried out for the PC to obtain the MAC address of the Web Server.
3. After obtaining the MAC address of the Web Server, the PC proceeds to send the HTTP request.
    a. What is the HTTP request method used in this scenario?
    b. If the server encounters an internal problem while processing the request, what HTTP response code would be returned to the PC?

Q 3(c) **[10 Marks]**

Explain how CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) is used to prevent collisions in the following scenario:
Computer A wants to send data to Computer D. Computer A is within the transmission range of Computers B and D, while Computer D is within the transmission range of Computers A and C.

[End of Question 3]

QUESTION 4 ***[TOTAL MARKS: 18]***

Answer the following multiple-choice questions by writing down the correct answer. Each question is worth 1.5 marks.

1. The main drawback of millimeter waves is that they:
   a) Provide low bandwidth
   b) Cannot penetrate obstacles
   c) Increase latency
   d) Reduce capacity

2. In the troubleshooting methodology, which step comes after identifying the problem?
   a) Implementing a solution
   b) Establishing a theory of probable cause
   c) Documenting the findings
   d) Verifying functionality

3. Which principle ensures no single person controls all aspects of a sensitive process?
   a) CIA Triad
   b) Separation of Duties
   c) Defense-in-Depth
   d) Authentication

4. What ensures that data remains accurate and trustworthy?
   a) Confidentiality
   b) Authentication
   c) Availability
   d) Integrity

5. What type of malware disguises itself as legitimate software?
   a) Worm
   b) Rootkit
   c) Trojan horse
   d) Spyware

6. DoS attack primarily aims to:
   a) Steal data
   b) Encrypt files for ransom
   c) Make resources unavailable
   d) Escalate privileges

7. Which cloud service model provides developers with tools and frameworks to build applications?
   a) IaaS
   b) PaaS

c) SaaS
d) FaaS

8. Which cloud deployment model shares infrastructure among multiple organizations with common goals?
a) Public cloud
b) Private cloud
c) Hybrid cloud
d) Community cloud

9. What is the role of the hypervisor?
a) Manage virtual machines and allocate resources
b) Encrypt network packets
c) Store data redundantly
d) Control Wi-Fi authentication

10. What is the biggest efficiency advantage of containers compared to VMs?
a) They include their own OS
b) They boot much faster and consume fewer resources
c) They cannot scale dynamically
d) They use circuit switching

11. What does NFV replace with software-based functions?
a) Virtual machines
b) User devices
c) Dedicated network appliances
d) Hypervisors

12. What is the core concept of IoT?
a) Static IP addressing
b) Using only 2.4 GHz frequency
c) Limiting cloud connectivity
d) Connecting physical objects to exchange data

[End of Question 4]

[END OF EXAM]